

SAYANEE HALDAR

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RESEARCH INTEREST

My study focuses on the multi-scale plasma processes that control space weather and the physics of solar–terrestrial interaction. In order to better constrain the connectivity between solar surface structures and heliospheric outflows, I am particularly interested in combining EUV remote sensing and in situ solar wind measurements with coronal magnetic field modeling. This builds on my experience with ionospheric irregularities, Alfvén wings, and solar-storm-driven instabilities. The creation and application of MHD simulations to reinterpret inner boundary conditions for global solar wind models is of concurrent interest. My goal is to enhance the physical realism and predictive power of heliospheric models by combining machine-learning techniques for data-driven detection and classification with physics-based modeling.

RELEVANT RESEARCH EXPERIENCE

- Alfvén wings in Earth's magnetosphere induced by solar storms

September 2024 - Present

Independent Work

- Analyzed magnetospheric plasma dynamics during major solar events using OMNI satellite datasets.
- Designed a machine learning framework (unsupervised clustering) to identify patterns during Alfvén wing formation events.
- Discovered and characterized outer Alfvén wings during the Mother's Day Storm (May 2024), contributing new insights into storm-time magnetospheric structure.

Tools: Python (SciPy, NumPy, Matplotlib), Machine Learning (Scikit-learn), Data Assimilation Techniques.

• Asymmetries in the Structural Features of Equatorial Ionization Anomaly

March 2024 - June
2024

Post Graduate Diploma Thesis, Supervisor: Dr. K. Venkatesh; PRL Ahmedabad

Ahmedabad, India

- Used a novel algorithm VISTA to investigate EIA events in Indian Equatorial Region
- Investigated EIA asymmetries in the Indian Equatorial Region using VISTA, a novel analysis algorithm.
- Conducted statistical analysis of ionospheric events linked to geomagnetic conditions.

Tools: MATLAB, Python

• Ionospheric Disturbances induced by Solar Flares in Indian Equatorial Region

May 2022 - August
2022

Summer Research Project, Supervisor: Dr. K. Venkatesh; PRL Ahmedabad

Ahmedabad, India

- Classified ionospheric perturbations associated with major solar flare events during 2021-2022.
- Developed data processing pipelines to extract and visualize disturbance patterns from GNSS data.

Tools: MATLAB, IRI Modeling.

OTHER RESEARCH EXPERIENCE

Stratification of Bay of Bengal and Climate Change

May 2025 – Sep 2025, Goa, India

Project Associate, Project Leader: Dr. Subhra Prakash Dey, National Institute of Oceanography

- Investigating stratification patterns in the Bay of Bengal under changing climate scenarios.
- Conducting statistical and machine learning-based analyses on reanalysis and CMIP6 datasets.
- Developing tools for spatiotemporal analysis of temperature, salinity, and density gradients.

Tools: Python, Julia

EDUCATION

PG Diploma in Space and Atmospheric Science

Physical Research Laboratory

Sep 2023 - May 2024

Ahmedabad, India

M.Sc Atmospheric Sciences

National Institute of Technology, Rourkela

Sep 2021 - Jun 2023

Rourkela, India

B.Sc (Physics, Mathematics, Geology)

Presidency University, Kolkata

Jul 2012 – Aug 2015

Kolkata, India

Conference Proceedings

- Halдар, S., (2023), **Seasonal trend of Antarctic Glacial Tsunami genesis impacted by decadal temperature increase**; 2nd Frontiers in Geosciences Research Conference, Physical Research Laboratory, Ahmedabad; Page 359
- Halдар, S., (2023), **Mars Ionospheric Disturbances induced by strong Solar Flares from January 2021–January 2023**; Indian Planetary Science Conference 2023, Physical Research Laboratory, Ahmedabad. Page 164
- Halдар, S. and Basu, S., 2024, January. **Ionospheric disturbances induced by consecutive solar eclipse and geomagnetic storms**. In *42nd meeting of the Astronomical Society of India (ASI)* (p. P247).
- Sayanee Halдар, (2025). **Alfvén Wing Identification in Mothers' Day Storm of 2024: Using OMNIData, 6th Indian Planetary Science Conference-2025**.

COURSES

- Deutsche Zertifikat A1 2019
- Deutsche Zertifikat A2 2020
- Deutsche Zertifikat B1 2021
- Deutsche Zertifikat B2 2022
- UAV REMOTE SENSING – TECHNOLOGICAL ADVANCES AND APPLICATION 2017
North Eastern Space Application Centre, ISRO, Department of Space, Govt. of India
 - Project on Applications of Remote Sensing in the field of Seismology: Case study from North Eastern Region (NER) of India
- BASICS OF REMOTE SENSING AND GEOGRAPHICAL INFORMATION SYSTEMS 2016
North Eastern Space Application Centre, ISRO, Department of Space, Govt. of India
 - Mini Project on Digital Elevation Model of Shillong Plateau

SEMINARS

- ISRO-Structured Training Program on "Sun-Planet Interactions: A Space Weather Perspective" 25th - 29th Sep 2023
Physical Research Laboratory Ahmedabad, India
 - **Topics on:** solar variability, space weather, planetary atmospheres, ionospheres, and magnetosphere and introduces important aspects of instrumentation and modelling techniques to investigate Sun-Planet interactions.
- Land-Atmosphere Interaction
National Institute of Technology, Rourkela

AWARD & FELLOWSHIP

AGU25 BERKNER TRAVEL FELLOWSHIP

September 2025

Awarded based on independent research on Alfvén waves.

PHYSICAL RESEARCH LABORATORY SUMMER RESEARCH FELLOWSHIP

May 2022

SKILLS

- **Programming Languages:** Python, MATLAB, Julia
- **Machine Learning:** Supervised and unsupervised learning, neural networks, data-driven modeling
- **Numerical Modeling:** Pluto, WRF
- **Tools & Libraries:** TensorFlow, PyTorch, NumPy, SciPy, Matplotlib, SunPy, AstroPy
- **Operating Systems:** Windows, Linux, macOS
- **Miscellaneous:** Scientific illustration, GitHub, LATEX, Blogging, MS Office, Ferret, GrADS

LANGUAGES

- **English** (Professional proficiency), **Bengali** (Native proficiency), **German** (Professional Proficiency), **Hindi** (Limited Working Proficiency), **French** (Elementary Proficiency)